

EDUCATION	The Chinese University of Hong Kong <i>Ph.D. in Computer Science and Engineering</i> • Advisor: Prof. Hong Xu • Research area: Machine Learning Systems Hong Kong SAR, China 2022.08 - 2026.07
	Northwestern Polytechnical University <i>B.E. in Computer Science and Technology</i> • GPA: 93.37/100, Rank: 1/247. • National Scholarship twice, First-class Scholarship three times, Outstanding Graduate of Northwestern Polytechnical University. Xi'an, China 2018.09 - 2022.06
RESEARCH INTEREST	I am broadly interested in systems design for machine learning workloads, especially the following topics: Distributed Machine Learning Systems: Optimizing distributed training for large models, including LLMs, video diffusion models, and RLHF/agent-RL workloads; designing and optimizing inference systems for machine learning workloads. Resource Management and Scheduling: Designing resource management, workload scheduling, and scheduling algorithms for machine learning systems.
INTERNSHIPS	Alibaba Star, RL Infra Group, Qwen Hangzhou, China • Contributed to performance optimization of RL infrastructure, focusing on resource scheduling, pipeline orchestration, and execution efficiency through system analysis and engineering optimizations. 2026.06 - 2026.08
	System Group, StepFun Beijing, China • Led research on large-scale video generation training efficiency. Designed and implemented DSV for dynamic attention sparsity in video DiT training, including low-rank approximation and two-stage sparsity prediction, sparse-attention kernels, sparse-aware context parallelism, and load-balancing strategies. The work was published at ASPLOS'26 and achieved up to 3.02x training throughput improvement on a 128-GPU cluster. • Characterized compute-network rhythms and resource bottlenecks across rollout, inference, and training components in large-scale RLHF/agent-RL post-training clusters. Designed dynamic compute-network orchestration and a reconfigurable fabric for RL clusters; built a simulation and evaluation framework supporting different model sizes, parallel strategies, cluster topologies, and workload rhythms. The work was accepted by SIGCOMM'26. 2024.09 - 2026.01
	Network Research Group, Microsoft Research Asia Remote • Developed a real-time, non-intrusive monitoring system to collect critical AI infrastructure metrics (GPU utilization, network/NVLink bandwidth), and designed a scalable analytics tool to evaluate ML workloads across large-scale clusters. • Analyzed six months of datacenter workload data to characterize resource usage and network patterns of various AI tasks, providing actionable recommendations to optimize cloud infrastructure and software stack. 2021.10 - 2022.03
PUBLICATIONS	Preprints - Kaiwen Chen, Xin Tan, Jingzong Li, Hong Xu, Libra: Efficient Resource Management for Agent-RL Post-Training. <i>Preprint</i>. [Preprint] - Yicheng Feng, Xin Tan, Yangtao Deng, Yimin Jiang, Yibo Zhu, Hong Xu, Frontier: Towards Comprehensive and Accurate LLM Inference Simulation. <i>Preprint</i>. [Preprint] [Code]

Peer-reviewed Publications

1. [SIGCOMM'26, CCF-A] Xin Tan, Yicheng Feng, Yu Zhou, Yimin Jiang, Yibo Zhu, Hong Xu, **Dynamic Compute and Network Orchestration for Disaggregated RL**. *ACM Special Interest Group on Data Communication (SIGCOMM)*, 2026. [\[Preprint\]](#)
2. [ASPLOS'26, CCF-A] Xin Tan, Yuetao Chen, Yimin Jiang, Xing Chen, Kun Yan, Nan Duan, Yibo Zhu, Daxin Jiang, Hong Xu, **DSV: Exploiting Dynamic Sparsity to Accelerate Large-Scale Video DiT Training**. *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2026. [\[Publication\]](#) [\[Preprint\]](#) [\[Code\]](#)
3. [ASPLOS'25, CCF-A] Xin Tan, Yimin Jiang, Yitao Yang, Hong Xu, **Towards End-to-End Optimization of LLM-based Applications with Ayo**. *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2025. [\[Publication\]](#) [\[Preprint\]](#) [\[Code\]](#)
4. Xin Tan, Yicheng Feng, Yu Zhou, Yimin Jiang, Yibo Zhu, Hong Xu, **RFabric: A Reconfigurable Network for the Rhythms of Disaggregated RL**. *The 1st Frontier AI Systems Workshop (FAISys)*, 2025. [\[PDF\]](#)
5. [ICPP'24, CCF-B] Xin Tan, Jiamin Li, Yitao Yang, Jingzong Li, Hong Xu, **Arlo: Serving Transformer-based Language Models with Dynamic Input Lengths**. *ACM International Conference on Parallel Processing (ICPP)*, 2024. [\[Publication\]](#)
6. [ICML'26, CCF-A] Kaihua Liang, Xin Tan, An Zhong, Hong Xu, Marco Canini, **FOCUS: DLLMs Know How to Tame Their Compute Bound**. *International Conference on Machine Learning (ICML)*, 2026. [\[Preprint\]](#) [\[Code\]](#)
7. [IWQoS'26, CCF-B] Kaiwen Chen, Xin Tan, Minchen Yu, Jingzong Li, Hong Xu, **MemShare: Memory Efficient Inference for Large Reasoning Models through KV Cache Reuse**. *IEEE/ACM International Symposium on Quality of Service (IWQoS)*, 2026. [\[Preprint\]](#)
8. Yicheng Feng, Xin Tan, Kin Hang Sew, Yimin Jiang, Yibo Zhu, Hong Xu, **Frontier: Simulating the Next Generation of LLM Inference Systems**. *SOSP Workshop on Practical Adoption Challenges of ML for Systems (SOSP-PACMI)*, 2025. [\[Preprint\]](#)

PROJECTS

Compute and Network Co-orchestration for Disaggregated RL

Accepted by SIGCOMM'26; first-author/lead work

2025.05 - 2026.01

- Systematically analyzed disaggregated rollout-training architectures for LLM RL post-training (RLHF/agent RL), identifying generation-stage bottlenecks caused by dynamic output lengths, KV-cache pressure, and long-tail requests, as well as phase-varying communication patterns across training, generation, and weight synchronization.
- Designed OrchestrRL, a compute-network co-orchestration framework with an adaptive rollout scheduler that dynamically switches among TP/EP/AFD parallel configurations based on batch size, generation-length distribution, and remaining request states, and mitigates stragglers through online request migration.
- Designed RFabric, a reconfigurable optical-electrical hybrid fabric that allocates bandwidth on demand for training collectives, generation communication, and weight synchronization; built RLSim, a high-fidelity simulator for evaluating RL clusters across model sizes, parallel strategies, cluster topologies, and workload rhythms.
- Validated OrchestrRL on a 64-GPU NVIDIA H800 cluster, achieving up to 1.42x end-to-end training throughput improvement; large-scale simulation shows RFabric approaches ideal non-blocking Fat-Tree performance and improves performance-cost efficiency by 1.53–2.06x.

High-fidelity Simulation for Modern LLM Inference Serving

Preprint; core contributor

2025.08 - 2026.05

- Analyzed simulation challenges as modern LLM serving evolves from homogeneous monolithic deployments to disaggregated architectures, where existing simulators struggle to jointly model prefill-decode disaggregation (PDD), attention-FFN disaggregation (AFD), complex parallelism, runtime optimizations, and stateful requests.
- Participated in designing and implementing Frontier, a discrete-event simulator that models co-location, PDD, and AFD through role-based cluster workers, incorporates runtime optimizations such as CUDA Graph, speculative decoding, and prefix caching into the scheduler-batch-engine loop, and supports stateful workloads such as reasoning, agents, and RL rollout.
- Contributed high-fidelity cost prediction modules for compute, communication, and memory, covering operator runtime, collective communication, KV-cache capacity, and cross-cluster transfer; Frontier achieved below 4% average throughput error in validation on a 16-GPU NVIDIA H800 cluster and scales to thousand-GPU simulations with SLA-aware design-space exploration.

Accelerating Large-scale Video Generation Training with Attention Sparsity

Published at ASPLOS'26; first-author/lead work

2024.09 - 2025.02

- Analyzed efficiency bottlenecks in large-scale video DiT training, showing that full attention incurs prohibitive cost for high-resolution long-video inputs while dynamic and heterogeneous attention sparsity offers substantial acceleration opportunities.
- Designed and implemented DSV, a training acceleration framework that combines low-rank approximation, two-stage sparsity prediction, and efficient sparse-attention kernels to predict critical key-value pairs and perform sparse computation.
- Extended and optimized context parallelism for sparse attention, including hybrid head/sequence parallelism, load balancing, and communication optimizations; implemented the system with PyTorch FSDP and Triton.
- Evaluated DSV on up to 30B-parameter video DiT models on a 128-GPU cluster, achieving up to 3.02x training throughput improvement while preserving generation quality compared with full attention.

End-to-End Optimization for LLM-based Applications

Published at ASPLOS'25; first-author/lead work

2024.01 - 2024.06

- Analyzed limitations of existing LLM application orchestration frameworks, including limited end-to-end optimization opportunities caused by modular orchestration and request scheduling that cannot directly optimize application-level objectives.
- Designed and implemented Ayo, an end-to-end optimization system centered on fine-grained orchestration, representing workflows across heterogeneous modules as dataflow graphs of task primitives to expose primitive characteristics and interactions for optimizations such as primitive parallelism and pipeline parallelism.
- Implemented Ayo on Ray and evaluated it on LLM applications including search-augmented generation and retrieval-augmented generation, achieving up to 2.09x average end-to-end latency speedup.

Serving Variable-length Language Model Workloads

Published at ICPP'24; first-author/lead work

2022.10 - 2023.06

- Analyzed inefficiencies in serving discriminative language models under variable-length requests, especially the limitations of static compilation with max-length padding and dynamic compilation.
- Proposed Arlo, a hybrid compilation and dynamic scheduling approach that serves requests of different lengths with corresponding statically compiled runtimes and dynamically adjusts GPU allocation according to request-length distributions.
- Designed a multi-level request queueing scheduler to handle short-term distribution shifts and optimize request latency; deployed and evaluated the system with Triton Inference Server and TensorRT, significantly improving efficiency under diverse request distributions.

AWARDS AND HONORS

- **Student Travel Grant**, ASPLOS 2025 2025.04
- **Full Postgraduate Scholarship**, CUHK 2022.08–2026.07
- **Outstanding Graduate**, Northwestern Polytechnical University 2022
- **First-class Scholarship**, Northwestern Polytechnical University 2021
- **First-class Scholarship**, Northwestern Polytechnical University 2020
- **National Scholarship**, Ministry of Education (China) 2020
- **National Scholarship**, Ministry of Education (China) 2019
- **First-class Scholarship**, Northwestern Polytechnical University 2019
- **Champion**, International Underwater Robot Competition, Simulation Track, Team Leader 2020.09
- **National Third Prize**, Loongson Cup National Undergraduate Computer System Design Contest, Team Leader 2020.06

SKILLS

Deep Learning/System Development: PyTorch, vLLM, Megatron, Verl, Triton, TensorRT, CUDA, NCCL
Programming: Python, C++, LaTeX
English: CET-4 619, CET-6 591, IELTS 7.0

ACADEMIC SERVICES

Reviewer: *IEEE Transactions on Network Science and Engineering*
Shadow Program Committee: *ACM EuroSys 2026*
Artifact Evaluation Committee: *USENIX OSDI/ATC 2025, USENIX OSDI/ATC 2024, USENIX FAST 2025, ACM CoNEXT 2025, ACM EuroSys 2025 Spring/Fall*

TEACHING

Teaching Assistant: *CSCI 3150, Introduction to Operating Systems, CUHK. 2023 Spring, CSCI 1120, Introduction to Computing Using C++, CUHK. 2022 Fall.*